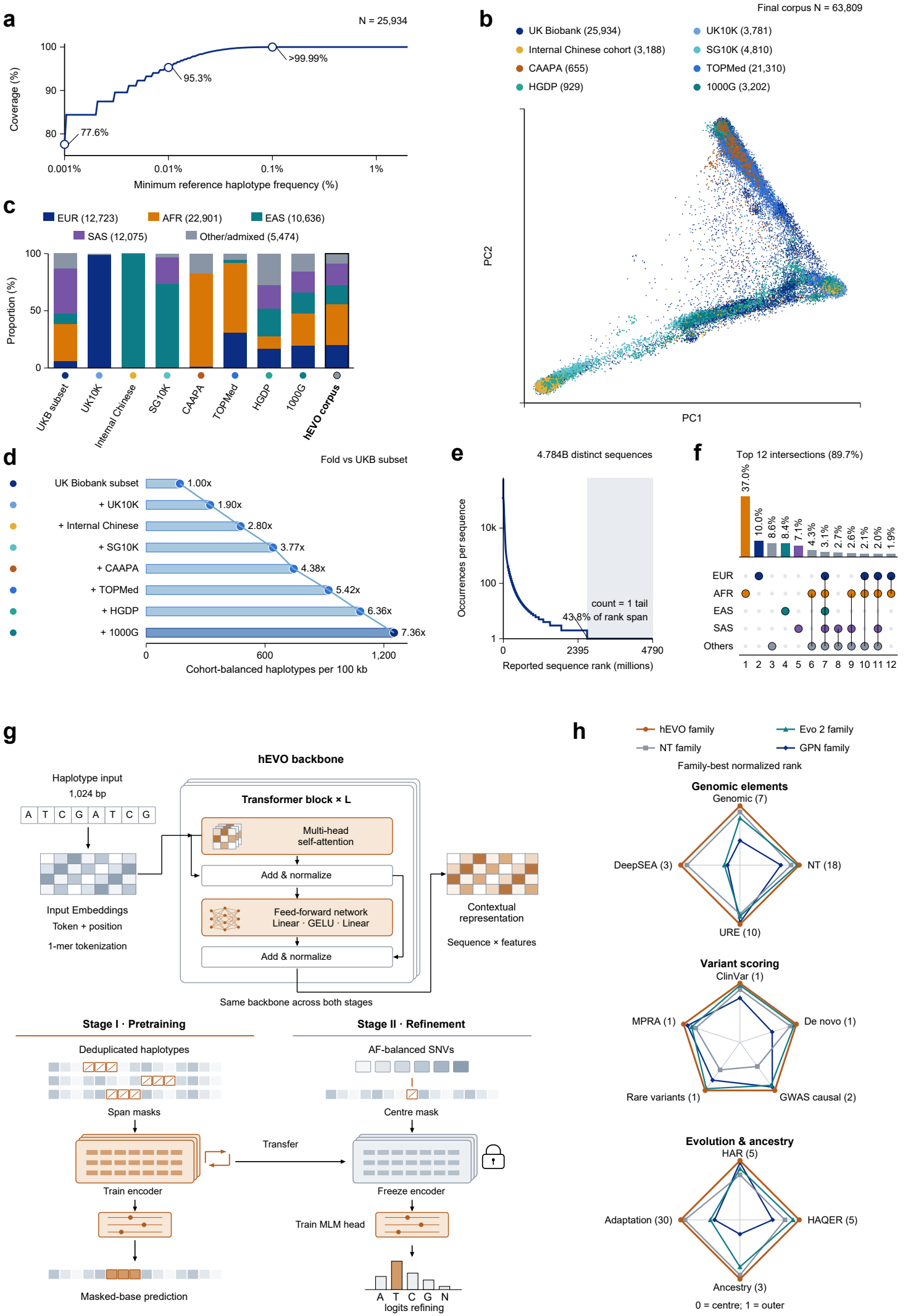

a
N = 25,934
Coverage (%)
100
90
80
77.6%
95.3%
>99.99%
0.001%
0.01%
0.1%
1%
Minimum reference haplotype frequency (%)
b
Final corpus N = 63,809
UK Biobank (25,934)
Internal Chinese cohort (3,188)
CAAPA (655)
HGDP (929)
UK10K (3,781)
SG10K (4,810)
TOPMed (21,310)
1000G (3,202)
PC2
PC1
c
EUR (12,723)
AFR (22,901)
EAS (10,636)
SAS (12,075)
Other/admixed (5,474)
Proportion (%)
100
50
0
UKB subset
UK10K
Internal Chinese
SG10K
CAAPA
TOPMed
HGDP
1000G
hEVO corpus
d
Fold vs UKB subset
UK Biobank subset
+ UK10K
+ Internal Chinese
+ SG10K
+ CAAPA
+ TOPMed
+ HGDP
+ 1000G
1.00x
1.90x
2.80x
3.77x
4.38x
5.42x
6.36x
7.36x
0
600
1,200
Cohort-balanced haplotypes per 100 kb
e
4.784B distinct sequences
Occurrences per sequence
10k
100
1
count = 1 tail
43.8% of rank span
0
2395
4790
Reported sequence rank (millions)
f
Top 12 intersections (89.7%)
37.0%
10.0%
8.6%
8.4%
7.1%
4.3%
3.1%
2.7%
2.6%
2.1%
2.0%
1.9%
EUR
AFR
EAS
SAS
Others
1 2 3 4 5 6 7 8 9 10 11 12
g
hEVO backbone
Haplotype input
1,024 bp
A T C G A T C G
Input Embeddings
Token + position
1-mer tokenization
Transformer block × L
Multi-head self-attention
Add & normalize
Feed-forward network
Linear · GELU · Linear
Add & normalize
Contextual representation
Sequence × features
Same backbone across both stages
Stage I · Pretraining
Deduplicated haplotypes
Span masks
Train encoder
Masked-base prediction
Transfer
Stage II · Refinement
AF-balanced SNVs
Centre mask
Freeze encoder
Train MLM head
A T C G N
logits refining
h
hEVO family
Evo 2 family
NT family
GPN family
Family-best normalized rank
Genomic elements
Genomic (7)
DeepSEA (3)
NT (18)
URE (10)
Variant scoring
ClinVar (1)
MPRA (1)
De novo (1)
Rare variants (1)
GWAS causal (2)
Evolution & ancestry
HAR (5)
Adaptation (30)
HAQER (5)
Ancestry (3)
0 = centre; 1 = outer